

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0208

Greens Bayou 69 138 kV
February 20, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0208. The Interconnection Customer proposes a 408.2 MW natural gas combined cycle facility interconnecting at the Greens Bayou 69 138 kV substation in the West Virginia area. The request was received on March 05, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 128 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,578,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
American Falls-Reata 138 kV	Greens Bayu 69-Kopperston 138 kV	180.9	221	100.0
Reata-Glenoveer 138 kV line	Greens Bayou 69-Kopperston 138 kV line	179.0	224	100.0
Greens Bayou 69-Glenoveer 138 kV	Kopperston-Greens Bayou 69 138 kV L	176.6	229	100.0
Kopperston-Greens Bayou 69 138 kV	Greens Bayou 69-Glenoveer Substation 138 kV	174.5	232	100.0
Keeler-Kopperston 138 kV	Glenoveer-Greens Bayou 69 138 kV	169.8	236	100.0
Oceana Substation-Keeler 138 kV	Glnoveer-Greens Bayou 69 138 kV	165.9	239	100.0
Oceana-Goodnoe 138 kV	Glenoveer-Greensbayou 69 138 kV line	159.7	245	100.0
Falconer-America Falls 138 kV Line	Scarbro-N Sweetwater 138 kV	153.1	191	14.9
Falconer-Pax Branch 138 kV L	N Sweetwater-Scarbro 138 kV	127.0	212	14.9
Pax Branch-Berlin 138 kV	N Sweetwater-Scarbro 138 kV	118.3	223	14.9
Scarbro-Berlin 138 kV	Scarbro-N Sweetwater 138 kV	115.8	226	14.9
Kean Mtn And Grassy Crk 138/1 kV TX	Kean Mtn And Grassy Crk-Clinch Riv 345 kV	109.7	345	8.3
Aspen 345/138 kV Transformer	Crandall-Aspen 345 kV Line	104.4	563	6.2
N Plains-Goodnoe 138 kV L	Glenoveer-Greens Bayou 69 138 kV L	103.2	244	66.9
N Plains Substation-Chauncey 138 kV line	Glenoveer-Greens Bayou 69 Substation 138 kV	101.5	246	66.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
American Falls-Reata 138 kV	Rebuild line with higher-capacity conductor	\$5,273,000	100% (\$5,273,000)
Reata-Glenoveer 138 kV	Rebuild line with higher-capacity conductor	\$3,777,000	100% (\$3,777,000)
Greens Bayou 69-Glenoveer Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,897,000	100% (\$3,897,000)
Greens Bayou 69-Kopperston 138 kV	Rebuild line with higher-capacity conductor	\$4,677,000	100% (\$4,677,000)
Keeler-Kopperston 138 kV	Rebuild line with higher-capacity conductor	\$5,760,000	100% (\$5,760,000)
Oceana Substation-Keeler 138 kV	Rebuild line with higher-capacity conductor	\$5,384,000	100% (\$5,384,000)
Goodnoe-Oceana 138 kV	Rebuild line with higher-capacity conductor	\$19,281,000	100% (\$19,281,000)
American Falls Substation-Falconer 138 kV	Rebuild line with higher-capacity conductor	\$3,645,000	28% (\$1,026,000)
Falconer-Pax Branch 138 kV L	Rebuild line with higher-capacity conductor	\$4,552,000	28% (\$1,281,000)
Berlin-Pax Branch 138 kV	Rebuild line with higher-capacity conductor	\$3,781,000	28% (\$1,064,000)
Berlin-Scarbro 138 kV line	Rebuild line with higher-capacity conductor	\$1,191,000	28% (\$335,000)
Kean Mtn And Grassy Creek 138/1kV Transformer Bank	Replace transformer with higher-rated unit	\$6,482,000	11% (\$714,000)
Aspen 345-138 kV Transformer	Replace transformer with higher-rated unit	\$25,589,000	5% (\$1,337,000)
N Plains-Goodnoe 138 kV L	Reconductor line	\$1,538,000	87% (\$1,345,000)
N Plains Substation-Chauncey 138 kV line	Reconductor line	\$1,632,000	87% (\$1,427,000)
	Total allocated to project		\$56,578,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.